Programming Fundamentals II: Lab 08

Due Date: Tuesday, Nov 05, 2024

Part 1. Sorted (10 Marks)

For this lab, we're going to practice working with sorting using the Comparable and Comparator interfaces. Download the Driver file provided. In this file, we instantiate an array of all of Canada's prime ministers. Each PM has a first name, last name, and an array of the dates they were elected. You may assume the PMs' elected dates are in order.
For their election dates, we use the LocalDate class:
https://docs.oracle.com/en/java/javase/22/docs/api/java.base/java/time/LocalDate.html

a. **(1 Mark)** Write the code for the PrimeMinister class. Your fields should be private. Give your PM a toString that displays their first name, last name, and years elected.
b. **(3 Marks)** We might say the most natural way to order PMs is by the dates they were **first elected.** Implement this natural ordering on your PM class. Modify your driver so it sorts the array of PMs and prints them out.
c. **(3 Marks)** We might often wonder "Who are the most recent PMs?" Modify your driver to display the 5 most recently elected PMs. To do this, you'll need to use an alternate ordering method that sorts PMs by their most recent elected dates.
d. **(3 Marks)** Another common way to sort people is by their names. Implement an alternate ordering of PMs to order them by their last names ascending. If two PMs have the same last name, break the tie by using their first names ascending. Modify your driver to display every PM elected before the year 1900 in this order.

Sample output on the following page.

Add the **date and your name and student number** to the top of your java files. Please submit your java files to Nexus before the end of the day on the lab date. **Late submissions will be not be accepted.**

Sample output:

```
Natural Sort:
      Sir John A. Macdonald: [1867-07-01, 1873-05-02]
      Alexander Mackenzie: [1873-05-02]
      Sir John Abbott: [1891-05-16]
      Sir John Thompson: [1892-10-05]
      Sir Mackenzie Bowell: [1895-12-21]
      Sir Charles Tupper: [1896-05-01]
      Sir Wilfrid Laurier: [1896-06-06, 1900-03-27]
      Robert Borden: [1911-10-10, 1915-10-24]
      Arthur Meighen: [1920-07-10, 1921-06-29]
      William Lyon Mackenzie King: [1921-09-06, 1925-01-01, 1935-01-30]
      Richard B. Bennett: [1930-10-07]
      Louis St. Laurent: [1943-11-15]
      John Diefenbaker: [1957-03-31, 1959-06-18]
      Lester B. Pearson: [1963-04-22, 1965-04-20]
      Pierre Trudeau: [1968-04-20, 1980-02-18, 1984-06-30]
      Joe Clark: [1979-06-04]
      Brian Mulroney: [1984-06-30, 1988-11-21]
      John Turner: [1984-08-03]
      Kim Campbell: [1993-06-25]
      Jean Chrétien: [1993-10-25, 1997-06-02, 2000-11-27]
      Paul Martin: [2003-12-12]
      Stephen Harper: [2006-01-23, 2008-10-14, 2011-05-02]
      Justin Trudeau: [2015-10-19, 2019-10-21, 2021-09-20]
5 Recent:
      Justin Trudeau: [2015-10-19, 2019-10-21, 2021-09-20]
      Stephen Harper: [2006-01-23, 2008-10-14, 2011-05-02]
      Paul Martin: [2003-12-12]
      Jean Chrétien: [1993-10-25, 1997-06-02, 2000-11-27]
      Kim Campbell: [1993-06-25]
Alphabetical before 1900:
      Sir John Abbott: [1891-05-16]
      Sir Mackenzie Bowell: [1895-12-21]
      Sir Wilfrid Laurier: [1896-06-06, 1900-03-27]
      Sir John A. Macdonald: [1867-07-01, 1873-05-02]
      Alexander Mackenzie: [1873-05-02]
      Sir John Thompson: [1892-10-05]
      Sir Charles Tupper: [1896-05-01]
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